H SAF DATA RETRIEVAL AND PILOT METHODOLOGY

Mnjoli catchment, Eswatini | Order 53877 | Sample date: 26 October 2026

The H SAF package arrived about three days after submission. All four archive checksums match. The sample is useful for learning the product formats and checking local coverage, but the rainfall products currently disagree and should not yet be treated as a definitive catchment rainfall total.

Purpose and scope

This report records the end-to-end H SAF sample retrieval, the files received, how the product metadata and spatial coverage were checked, and what the first Mnjoli-area comparison shows. It is an acquisition and data-quality note, not a validation study, water-balance result, or operational forecast.

At a glance

Item	Evidence / result
Order	Four products (H64, H90, H14, H122), requested for 2026-10-26; portal reference 53877.
Processing time	About 3 days from the portal order timestamp on 8 Nov to the availability notification on 11 Nov. Exact elapsed time is approximate because the portal timestamp timezone was not stated.
Package	Notification stated 426 MB. The four downloaded tar.gz archives total 446,325,321 bytes (about 426 MiB); the archives are compressed.
Integrity	All four archives matched the supplied MD5 manifest. They were extracted locally; no source code was changed.
Main unresolved result	H64 reports 0.0 mm at its single in-catchment grid-cell centre, while H90 reports about 7.48 mm mean over 12 in-catchment pixel centres for its 00Z 24-hour file. Confirm the H90 accumulation window before comparing.

Status: H SAF files were downloaded and inspected. An email asking H SAF to clarify the H90 24-hour time window and H64/H90 difference has been drafted, but not sent as of the report date.

1. Retrieval chronology

The acquisition moved from product selection through a queued order to FTP delivery. The portal reported the order in queue before the download notification arrived.

Date / stage	What happened
Account and product interests	The H SAF profile was set to precipitation, soil moisture, product downloads, and newsletter; snow was not selected.
Sample selection	A one-day common date, 2026-10-26, was chosen for the available H64, H90, H122, and H14 products. This was a global product order, not a catchment-clipped extraction.
Size check and confirmation	The draft estimate was H64 1 MB, H90 107 MB, H14 2 MB, H122 411 MB, total 521 MB. The user explicitly confirmed submission.
Order submitted	Portal order list showed reference 53877, submitted 2026-11-08 at 18:07:59 (portal display; timezone not explicit), status ?Order in queue.?
Wait and follow-up	The queue status persisted at checks on 9 and 10 November. A status email was sent on 9 November. The H SAF login later returned HTTP 500, preventing an online status check.
Availability notice	On 11 November, H SAF notified that the files were ready on the FTP server, with a stated total of 426 MB and a download expiry of 18 November 2026 at 15:50:50 UTC.
Transfer	FTP URLs timed out in browser attempts and programmatic transfers. WinSCP 6.7 was installed; the user transferred files into data/external/hsaf_order_53877. Credentials are intentionally not reproduced here.
Local verification	The five supplied files (four archives and the MD5 list) were found. All four tar.gz archives passed MD5 validation, then were safely extracted into product-specific subfolders.

Processing-time estimate: approximately 72 hours (about 3 days). This is derived from the order-list timestamp and the reported availability date/time; no timezone was visible beside the portal?s order timestamp, so a minute-precise duration cannot be asserted.

Acquisition and local organization

The retained package is in data/external/hsaf_order_53877. Original archives, the checksum manifest, and extracted files are kept together so the data can be rechecked or reprocessed. The directory is untracked by Git; it has not been committed.

Archive contents

Product archive	Downloaded archive bytes	Extracted data files	Observed formats
h122_01.tar.gz	330,760,798	960	NetCDF (.nc), ASCAT observations
h14_01.tar.gz	2,369,717	1	GRIB compressed with bzip2 (.grib.bz2)
h64_01.tar.gz	706,326	1	NetCDF compressed with gzip (.nc.gz)
h90_01.tar.gz	112,488,480	28	NetCDF compressed with gzip; 24 hourly and four 24-hour files
Total	446,325,321	990	Plus four tar directory entries

2. Product retrieval and file interpretation

The descriptions below combine the order-form product names with the attributes inside the delivered files. They describe what this sample contains; detailed algorithm settings should be taken from the matching H SAF product user manuals and version documentation.

ID / ordered product	How to interpret this sample and its file evidence
H90 - P-AC-SEVIRI_E	The order description is accumulated precipitation at ground from blended microwave/infrared observations over the IODC. Delivered NetCDF metadata identifies an hourly accumulated rain amount, a geostationary projection, 3712 x 3712 grid, satellite MSG2, and precipitation variable acc_rr in mm with 0.1 scale factor. The archive has 24 nominal hourly files and four 24-hour files (00, 06, 12, 18 UTC). H90 is the finer spatial sample in this order; 12 pixel centres lie inside the catchment boundary. The delivered metadata does not, by itself, settle the exact start/end window for each 24-hour filename.
H64 - P-AC-SM2RAIN	The delivered NetCDF title is ?24 hours Soil Moisture-PMW integrated rainfall?; the grid is regular 0.25 degree, and acc_rr is daily accumulated precipitation in mm. Its metadata explicitly gives end_of_accumulation_time=20261026T00:00Z and accumulation_time=24 hours. The file is a daily integrated precipitation estimate, not a point gauge. Only one grid-cell centre falls within the catchment.
H122 - ASCAT SSM NRT 6.25 km	The NetCDF global attributes identify a Metop-B ASCAT near-real-time surface soil-moisture product, product level 2. It contains swath observations, time, geolocation, surface_soil_moisture in percent saturation, uncertainty/noise, and surface/processing flags. Values carry a 0.01 scale factor; flags and missing-value codes must be applied before using a reading. This is a set of satellite retrieval observations, not a daily catchment raster mean.
H14 - SM-DAS-2	The delivered GRIB contains four fields named swi1-swi4, with units dimensionless, on a reduced Gaussian grid. The order description identifies a root-region soil wetness profile index retrieved through an ASCAT surface-wetness assimilation method. Treat these as model/assimilation-derived layer indices, not direct volumetric soil-water measurements. No grid-point centre falls inside the catchment; the nearest valid point is about 1.62 km outside it.

Processing performed locally

1. Listed archive members and checked for path traversal before extraction.
2. Compared each archive MD5 with the supplied order manifest using the Windows Get-FileHash implementation; all four hashes matched.
3. Inspected NetCDF global/variable attributes, dimensions, coordinates, units, scale factors, missing values, and flags. Decompressed H64/H90 gzip members and H14 bzip2 data for inspection.
4. Compared product-grid centres with the Mnjoli catchment GeoJSON polygon. For H90, grid scan-angle coordinates and the geostationary projection were transformed to geographic coordinates before testing pixel centres.
5. Summarized only valid, unmasked pixel values for the catchment; retained native-resolution counts and did not blend products with different methods/resolutions.

Quality handling

H122 includes missing values and processing flags; one in-catchment observation is filled/missing and is excluded. H90 includes qind (pixel quality index); the catchment-average qind values for the 24-hour files range from about 86 to 93 percent. These quality indicators do not resolve the H64/H90 algorithm or period difference. H14 values are adjacent-grid context because the nearest point is outside the polygon.

3. Mnjoli coverage and first comparison

The spatial test used the project catchment polygon in data/external/mnjoli_w60e_catchment.geojson. Its approximate bounds are 31.6083-31.7129 E and 26.0500-26.2250 S. Grid-centre counts matter: catchment averages from one or a dozen cells should not be mistaken for dense in-situ observations.

Product / file window	Spatial support at Mnjoli	Sample result	Interpretation
H64, 24 h ending 26 Oct 00:00 UTC	1 cell centre inside, 0.25 degree grid	0.0 mm at cell centre	Very sparse spatial support. The period end is explicit; start is inferred from 24-hour accumulation duration.
H90, 24 h file timestamped 26 Oct 00Z	12 pixel centres inside	7.48 mm mean; range 6.4-8.6 mm; mean qind 92.75%	Timestamped file window not confirmed from delivered metadata. Do not assert it matches H64 until H SAF confirms.
H90, 1 h files labelled 00-23 UTC on 26 Oct	12 pixel centres inside in each file	Sum of the 24 nominal hourly file means = 6.01 mm; rain mainly 13-15Z	A nominal-label sum, not an independent gauge validation. Confirm interval labeling if exact daily totals matter.
H90, other 24 h files at 06/12/18Z	12 pixel centres inside each	Means: 7.48 / 7.51 / 6.98 mm	These are overlapping windows, not additive. Do not sum 24 h products.
H122, valid readings on 26 Oct	One grid location inside	42.24%, 35.38%, 32.65% saturation; noise 3.64%, 3.91%, 3.56%	Three valid swath observations across the day at the same grid location; one additional in-polygon observation is missing/flagged. Not a catchment mean.
H14, four soil-wetness layers	No cell centre inside; nearest valid centre ~1.62 km outside	SWI1 0.402; SWI2 0.496; SWI3 0.645; SWI4 0.382 (dimensionless)	Nearby model/assimilation context only; not an in-catchment retrieval.

Rainfall conclusion: H64 and H90 conflict for what may be the same trailing 24-hour period (0.0 mm versus 7.48 mm). H90?s 00Z file has not been independently proven to cover the prior 24 hours. Keep both source estimates separate and unresolved; do not average them or convert them into a reservoir volume.

Why the estimates may differ

Potential factors include different retrieval inputs and algorithms, substantially different grid support (one H64 cell versus 12 H90 pixels), different representativeness within the catchment, quality screening, and possibly different accumulation windows. The sample alone cannot determine which factor explains the mismatch. H90 reports a spatial mean; H64 is a single grid-centre value. These are not equivalent catchment sampling designs.

Independent context is currently missing

The local Mnjoli CHIRPS and ERA5-Land daily series in the repository end on 30 December 2025, so they cannot validate 26 October 2026. The Open-Meteo ERA5-Land archive endpoint returned null precipitation values for 24-27 October 2026 during the check. The CHIRPS data host timed out during access. Therefore, no independent rainfall observation for the sampled date is included in this comparison.

4. Current assessment and next steps

This order demonstrates that H SAF can provide usable precipitation and soil-wetness files for Eswatini, but the sample also exposes practical limits: asynchronous order fulfilment, FTP-only delivery for this order, non-identical product grids, and ambiguous accumulation semantics that require product documentation or provider confirmation.

What is established

- Order 53877 included H64, H90, H14, and H122 for 26 October 2026.
- H SAF provided the package after approximately three days; the notification gave a seven-day download window ending 18 November 2026 at 15:50:50 UTC.
- All four downloaded tar.gz archives match the supplied MD5 checksums.
- H64 has one grid centre inside the catchment; H90 has 12; H122 has three valid observations at one in-catchment grid location; H14 has no grid centre inside the boundary.
- H64 and H90 provide materially different rainfall estimates in the first apparent overlapping 24-hour comparison.

What remains unresolved

- Whether H90?s 00Z 24-hour product is the trailing interval ending at 00Z, and whether it is directly time-aligned with H64.
- The reason for the H64/H90 discrepancy and the appropriate quality-screening rules for the specific product versions.
- Whether H90 hourly values? nominal timestamp labels correspond to intervals ending or beginning at each hour.
- Whether the point/pixel coverage is adequate for any operational Mnjoli rainfall or soil-moisture inference.

Recommended next actions

1. Send the prepared clarification email to H SAF; request exact H90 start/end times for the 00Z 24-hour file, the H90 hourly time-label convention, and guidance on H64/H90 comparisons and quality flags.
2. Retain the raw H SAF files and checksums; do not alter native values or discard flags.
3. Obtain an independent, date-matched rainfall observation or dataset before interpreting the one-day rainfall discrepancy. Compare like-for-like periods and document each source?s spatial support.
4. Evaluate catchment sampling over multiple dates and seasons. A one-day pilot cannot establish bias, skill, or suitability for a 30-day water-balance forecast.
5. Keep satellite precipitation and soil-moisture indicators separate from storage/water-balance targets until catchment geometry, units, time windows, and independent validation are established.

Data governance and use

H SAF states its products are provided without charge under CC BY 4.0, with attribution required. Keep product identifiers, version/creation metadata, and the download date with any derivative dataset or figure. Do not publish FTP credentials; they are not needed to cite or describe the data.

References and evidence

[1] H SAF official site and product information: hsaf.meteoam.it (accessed 11 Nov 2026). Product names and order settings are also recorded in portal order 53877.

[2] H SAF order 53877 delivered-file metadata: H90 NetCDF global/variable attributes; H64 NetCDF global/variable attributes; H122 NetCDF global/variable attributes; H14 GRIB message keys. Local copies are in data/external/hsaf_order_53877/extracted/.

[3] H SAF package MD5 manifest: data/external/hsaf_order_53877/hsaforder_53877.md5. All four archive hashes verified locally on 11 Nov 2026.

[4] H SAF data licence/attribution notice: Creative Commons Attribution 4.0 International (as stated on the H SAF site).

[5] Project catchment geometry: data/external/mnjoli_w60e_catchment.geojson. Independent reference files: data/external/mnjoli_chirps_daily_2015_2025.csv and data/external/mnjoli_era5_land_daily_2015_2025_processed.csv (both end 30 Dec 2025).

Prepared from portal observations, user-provided H SAF delivery notification, and locally inspected files. This report does not reproduce account or FTP credentials.